

TEACHING EXAM

Subject – MATHS

Topic – Percentage

LECTURE NO -01



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Topics to be Covered

Topic

Percentage



$$200\% \longrightarrow 90$$

$$1\% \longrightarrow \frac{90}{200}$$

$$50\% \longrightarrow \frac{90}{200} \times 50$$

$$22.5$$

What will be 50% of a number whose 200% is 90?
 उस संख्या का 50% क्या होगा जिसका 200%, 90 है?

- ☒ a) 22.5
- b) 22
- c) 36
- d) 34

What will be 50% of a number whose 350% is 70?

उस संख्या का 50% क्या होगा जिसका 350%, 70 है?

$$350\% \text{ — } 70$$

$$1\% \text{ — } \frac{70}{350}$$

$$50\% \text{ — } \frac{70}{350} \times 50$$

- a) 35
- ☒ b) 10
- c) 7
- d) 15

**What will be 200% of a number
whose 150% is 60?**

**उस संख्या का 200% क्या होगा जिसका
150%, 60 है?**

- a) 20**
- b) 40**
- c) 60**
- d) 80**

What will be 45% of a number whose 90% is 184?

उस संख्या का 45% क्या होगा जिसका 90%, 184 है?

- a) 94
- b) 48
- c) 92
- d) 46

→ If 32% of a number exceeds 27% of the same by 28, the number is ?
 एक संख्या का 32% उसके 27% से 28 ज्यादा है तो संख्या क्या होगी ?

- a) 560
- b) 360
- c) 460
- d) 720

$$32\% - 27\%$$

$$= 5\% \longrightarrow 28$$

$$N = 100\% \longrightarrow \frac{28}{5} \times 20 = 560$$

If 20% of a number exceeds 12% of the same by 48, the number is ?

एक संख्या का 20% उसके 12% से 48 ज्यादा है तो संख्या क्या होगी ?

a) 500

✓ b) 600

c) 400

d) 300

$$8\% \longrightarrow 48$$

$$1\% \longrightarrow 6$$

$$\begin{array}{r} 100\% \longrightarrow \\ \times 6 \quad 600 \end{array}$$

If 35% of a number exceeds 23% of the same by 600, the number is ?

एक संख्या का 35% उसके 23% से 600 ज्यादा है तो संख्या क्या होगी ?

$$12\% \longrightarrow 600$$

$$1\% \longrightarrow 50$$

$$100\% \xrightarrow{\times 50} 5000$$

a) 3000

b) 6000

c) 4000

d) 5000

Relationship b/w % & fractions

$$\text{percentage} = 100\% = \frac{100}{100} = 1 = \text{fraction.}$$

$$\frac{1}{2} \times 100\% \longrightarrow 50\%$$

$$\frac{1}{3} \times 100 = 33.33\% \text{ या } 33\frac{1}{3}\% \quad ; \quad \frac{2}{3} \rightarrow 66.66\%$$

$$\frac{1}{4} \times 100 = 25\% \quad ; \quad \frac{2}{4} = \frac{1}{2} = 50\% \quad \frac{3}{4} \times 100\% = 75\%$$

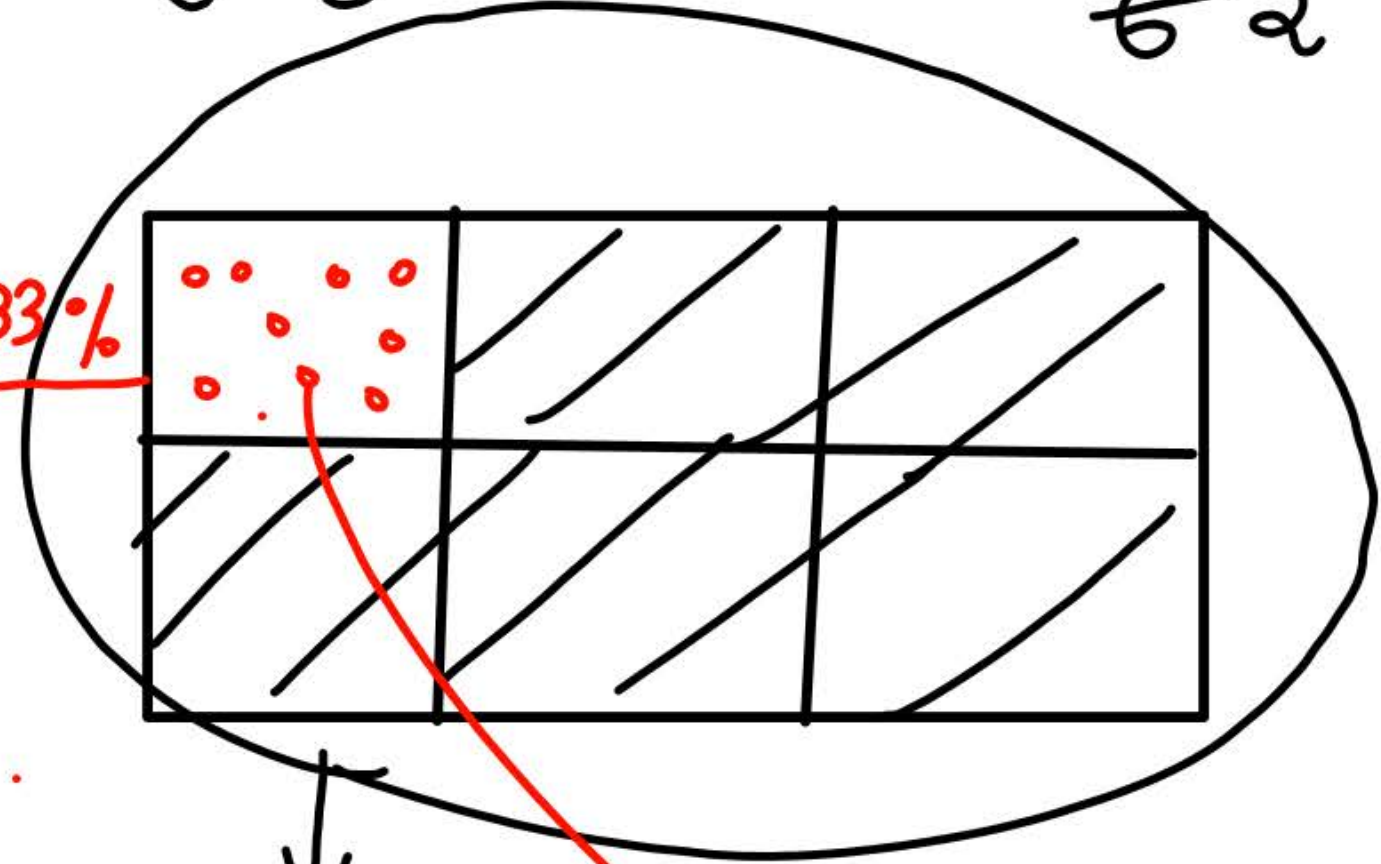
$$\frac{1}{5} \times 100 = 20\% \quad \frac{2}{5} = 40\% \quad \frac{3}{5} \rightarrow 60\% \quad \frac{4}{5} \rightarrow 80\%$$

$$\frac{1}{6} \times 100 = 16.66\% \text{ या } 16.67\% ; \frac{2}{3} = 33.33\% ; \frac{3}{2} = 50\%$$

या $16\frac{2}{3}\%$

$$\frac{2}{3} \rightarrow 66.66\%$$

$$\frac{5}{6} \rightarrow 83.33\%$$



$$100\% - 16.67\%$$

14 table
 $\frac{1}{7} \times 100 = 14.28\%$

$$\frac{2}{7} \rightarrow 28.56\% \quad ; \quad \frac{3}{7} \rightarrow 42.84\%$$

$$7 \sqrt{100 \overline{) 14.28}} \%$$

$$\begin{array}{r} -71 \\ \hline 30 \\ -28 \\ \hline 20 \\ -14 \\ \hline 60 \\ -56 \\ \hline 4 \end{array}$$

$$\frac{4}{7} \rightarrow \begin{array}{r} 56. \\ + 1.12 \\ \hline 57.12\% \end{array}$$

$$\frac{5}{7} \rightarrow \begin{array}{r} 70. \\ + 1.40 \\ \hline 71.40\% \end{array}$$

$$\frac{1}{8} \times 100\% = 12.5\%$$

$\frac{1}{4} \times \frac{1}{2} = 25\%$

$$\frac{2}{8} \times \frac{1}{4} = 25\%$$

$$\frac{3}{8} \rightarrow \begin{array}{r} 36 \\ + 1.5 \\ \hline 37.5\% \end{array}$$

$$\frac{4}{8} \times \frac{1}{2} = 50\%$$

$$\begin{array}{r} 12.5 \\ \times 3 \\ \hline 37.5 \end{array}$$

$$\frac{5}{8} \rightarrow \begin{array}{r} 60 \\ + 2.5 \\ \hline 62.5\% \end{array}$$

$$\frac{6}{8} \times \frac{3}{4} = 75\%$$

$$\frac{7}{8} \rightarrow \begin{array}{r} 84 \\ + 3.5 \\ \hline 87.5\% \end{array}$$

$\frac{1}{9}$ $\xrightarrow{\text{11 table}}$ $\times 100 = 11.11\%$ $\frac{2}{9} \rightarrow 22.22\%$; $\frac{3}{9} \rightarrow 33.33\%$; $\frac{4}{9} \rightarrow 44.44\%$
 $\dots$

$\frac{1}{11}$ $\xrightarrow{\text{9 table}}$ 09.09% $\frac{2}{11} \rightarrow 18.18\%$, $\frac{3}{11} \rightarrow 27.27\% \dots$

$$\begin{array}{r} 9 \overline{) 100} \text{ (11.11)} \\ \underline{-91} \\ 100 \\ \underline{-91} \\ 10 \end{array}$$

$10 - 9 = 10 \div 9 = 1$

2 minutes Summary

Topic

Percentage



THANK YOU